Justin Mascarenhas

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Summary

Computer engineer (M.S. Computer Engineering, RIT, 2026) who builds the firmware between the sensor and the model: register-level bare-metal C on Cortex-M, interrupt-driven motor and servo control, and the Linux services that tie microcontrollers to NVIDIA Jetson edge compute.

Skills

Embedded & Firmware: Bare-metal C (STM32 Cortex-M4), Arduino (AVR, Due), FreeRTOS, Interrupts & timers, PWM motor & servo control, Quadrature encoders, UART / I2C / SPI, ADC
Embedded Linux & Edge: NVIDIA Jetson, Raspberry Pi, systemd services, Static IP networking, TensorRT, CUDA
Hardware & Debug: Oscilloscopes, Logic analyzers, EAGLE PCB design, Proteus, Multimeters
Languages & Tools: C/C++, Python, Bash, Linux, Git, CMake

Experience

Embedded Systems Intern | **Planctech Innovations** Jul 2022 – Nov 2022
- Engineered a wearable gesture-recognition glove for hands-free cursor control, streaming 3-DoF accelerometer motion from a Raspberry Pi over I2C and Bluetooth into HID commands.

Research Intern (Capstone) | **Bhabha Atomic Research Centre (BARC)** Jul 2023 – Apr 2024
- Architected a metal-surface inspection pipeline for microscopic crack analysis that combines custom image stitching, wavelet denoising and depth estimation.

General Secretary | **Agnel Robotics Club (ARC), F.C.R.I.T.** Jul 2023 – May 2024
- Designed and fabricated single-layer PCBs in EAGLE for locomotive robots, with UART and I2C sensor communication and closed-loop feedback.

Projects

Voice-Controlled Robotic Arm (COFFIN) — M.S. Capstone, RIT Apr 2026
- Implemented two-link inverse kinematics in C++ on the Arduino, driving five servos with contact-aware grasping from dual time-of-flight sensors on an I2C multiplexer: a camera range gate plus a gripper that closes at 25 mm.
- Deployed the Jetson arm bridge as a systemd service (auto-restart with backoff, non-root isolation, environment-based config) on static-IP networking for reliable multi-device WebSocket registration.
- Drove the system to 100% pick reliability (10/10 trials) at ˜11.5 s voice-to-pick by hardening it against false-positive detections, servo brown-out and stale-frame capture.

STM32 Bare-Metal Temperature Monitor
- Engineered Cortex-M4 temperature-monitoring firmware in bare-metal embedded C, with timer-synchronized sensing and a real-time display and no HAL or RTOS layer.
- Programmed every peripheral at the register level (GPIO, TIM2, ADC1, I2C1): timer-triggered ADC scan conversions, interrupt-driven sample capture, and a hand-written I2C driver.
- Fit the complete firmware in 3.68 KB Flash / 1.58 KB RAM at a deterministic 1 Hz update rate; verified in Proteus.

Object Tracking Turret
- Built a real-time tracking turret: YOLOv8 on an NVIDIA Jetson detects and tracks targets and streams control commands over UART to STM32 firmware.
- Wrote multi-servo PWM control driven by interrupt-driven UART command handling, with a manual user override.
- Coordinated sensing, control and display tasks under FreeRTOS with mutex-protected shared memory, adding time-of-flight range sensing and a live OLED distance readout.

Robocon Competition Robot — Holonomic Drive & Encoder Odometry 2022 – 2024
- Built closed-loop heading correction for a four-wheel holonomic robot on Arduino Due, deriving heading from dual-axis quadrature-encoder odometry (2400 CPR) and feeding the angular error back into rotation PWM.
- Implemented interrupt-driven quadrature decoding from first principles as a 4-bit A/B transition state machine, and drove four motor channels at 15 kHz PWM.

Education

M.S., Computer Engineering | Rochester Institute of Technology Aug 2024 – May 2026
Courses: Multiple Processor Systems, High Performance Architecture, Principles of Robotics, Robot Perception

B.E., Electronics & Telecommunication | University of Mumbai (F.C.R.I.T.) 2020 – 2024
Courses: Digital Electronics, Digital Signal Processing